Rosseland-Mean Opacity of Pure Hydrogen Gas under LTE

Benchmarking a Continuum-Only Model against LANL/TOPS

Zisen Dai

ASTRON C207: Radiation Processes in Astronomy · UC Berkeley · Spring 2026

Goal: compute $\kappa_{\mathrm{R}}(T, \rho)$ for pure H in LTE and benchmark a continuum-only model against LANL/TOPS over $T \in [5 \times 10^{-4},\ 10]$ keV and $\rho \in [10^{-12},\ 10^{-3}]\ \mathrm{g\,cm^{-3}}$.

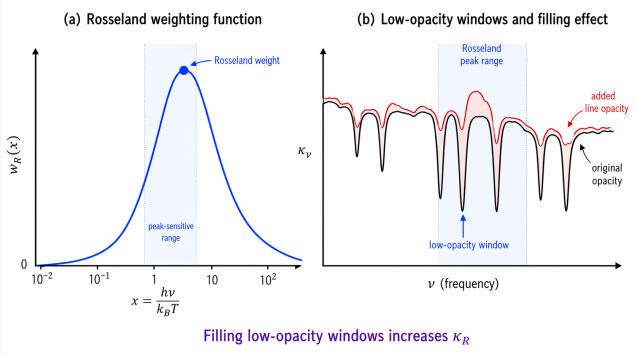
Physical Background: What Is the Rosseland-Mean Opacity?

The Rosseland mean is the **appropriate gray opacity for LTE diffusion**, weighted by $\partial B_\nu / \partial T$:

$$\frac{1}{\kappa_{\rm R}} = \frac{\int_0^\infty \left(\kappa_\nu^{\rm tot}\right)^{-1} \frac{\partial B_\nu}{\partial T}\, {\rm d}\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T}\, {\rm d}\nu}$$

$$\frac{\partial B_\nu}{\partial T} \propto w_R(x) = \frac{x^4 e^x}{(e^x - 1)^2}, \qquad x \equiv \frac{h\nu}{k_B T}$$

- **Harmonic mean:** $\kappa_{\rm R}$ is dominated by *low-opacity windows*; one transparent channel suppresses $\kappa_{\rm R}$ far more than any opaque region raises it.
- **Consequence:** missing line opacity leaves a **strong imprint on** $\kappa_{\rm R}$, even the EOS is approximately correct.



(a) $w_R(x)$ peaks near $x \approx 3.8$; the harmonic mean is dominated by low-opacity windows.

(b) Line opacity fills those windows and raises $\kappa_{\rm R}$ significantly.

Final Production Model and Benchmark Scope

Included physics

- Klein–Nishina electron scattering
- e–p free-free absorption
- Neutral-H bound-free photoionization
- H^- bound-free absorption
- H^- free-free absorption (John 1988)
- Density-dependent level truncation
- Capped ionization-energy lowering

Excluded physics

- × Bound-bound (line) opacity
- × Non-LTE effects
- × Smooth Hummer–Mihalas occupation probabilities

Formal Benchmark Domain

$T \in [5\times10^{-4},\ 10]$ keV

$\rho \in [10^{-12},\ 10^{-3}]$ g cm^{-3}

Reference: LANL/TOPS pure-H tables

Overall Benchmark Score

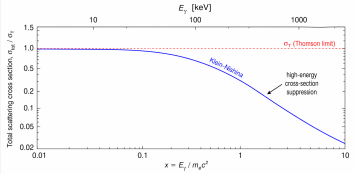
238 / 276 grid points within 10% of TOPS

86% overall pass rate

Key fixes: cold/dense corner → H^- free-free | hot-T bias → Klein–Nishina | dense-corner EOS → capped lowering

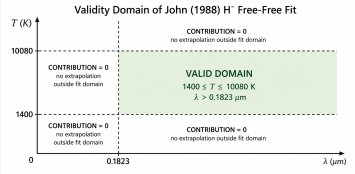
Three Implementation Choices That Control the Benchmark Behavior

(1) High-T Scattering



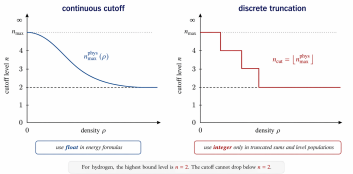
- Use σ_{KN} instead of σ_T
- Taylor branch at $x \ll 1$ for stability
- Removes ~15% overestimate above 3 keV

(2) Cold/Dense Continuum



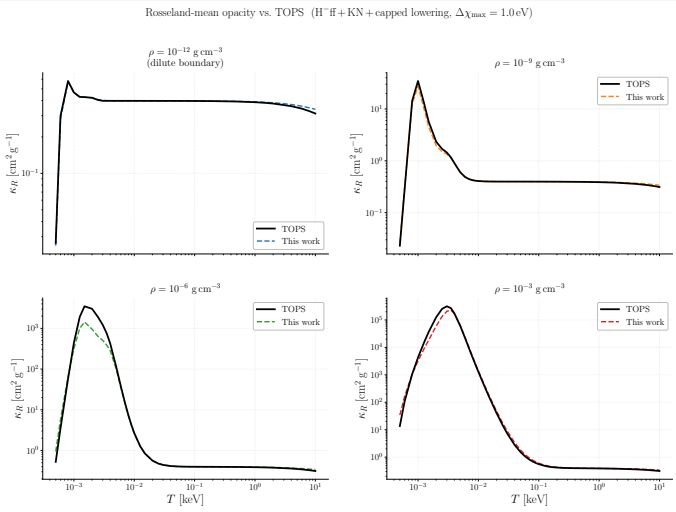
- H^- free-free: John (1988) fit
- Valid $T \in [1400, 10080]$ K; zeroed outside
- >100× error without it in cold/dense corner

(3) Capped Lowering & $n_{\max}$



- Float $n_{\max}^{\mathrm{phys}}$ in energy formulas; n_{cut} only for partition sums
- $\Delta\chi_{\mathrm{raw}} = 13.6(\frac{1}{n^2} - \frac{1}{n_{\max}^2})$ eV
- $\Delta\chi_{\mathrm{eff}} = \min(\Delta\chi_{\mathrm{raw}},\ 1.0\,\mathrm{eV})$
- Full lowering: ~16× over-ionization; cap: ~1.3×

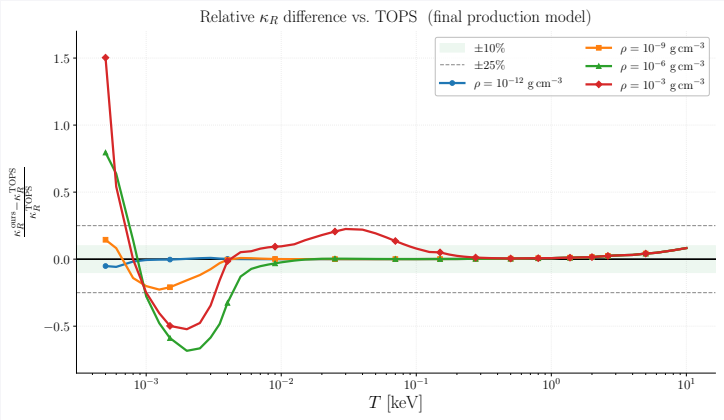
Global Benchmark: Rosseland-Mean Opacity vs. LANL/TOPS



- Final model reproduces the **overall TOPS opacity structure** across all four benchmark densities.
- Agreement is **excellent** in the relatively dilute regime and at $T \gtrsim 10$ mkeV.
- Residual mismatch is **localized**, not global — the dominant continuum physics is already captured over most of the formal domain.

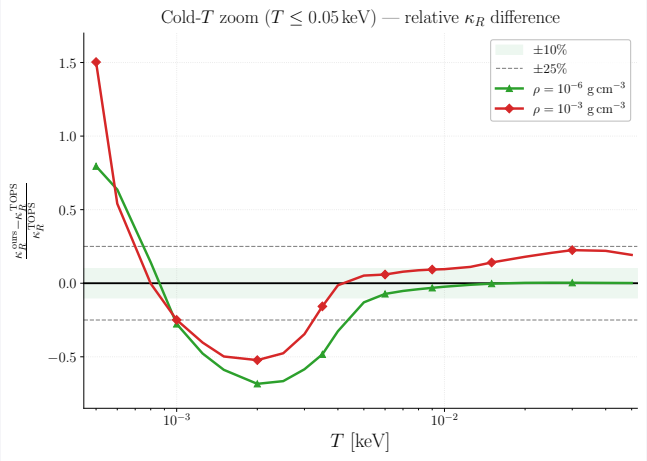
Strongest residuals are confined to low-to-intermediate T at higher ρ.

Residual Structure: Where Does the Model Still Fail?

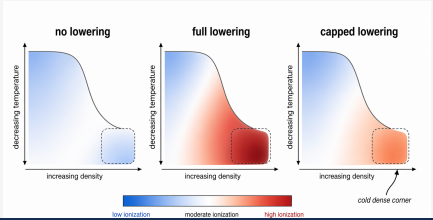


- **Hot regime:** largely repaired by Klein–Nishina; <5% error above 10 mkeV.
- **Partially neutral** (ρ=10^{-6} g cm^{-3}): persistent −30 to −50% deficit — the dominant remaining discrepancy.
- **Cold dense corner** (ρ=10^{-3} g cm^{-3}): additional sensitivity at lowest T.
- Residuals are **regime-specific**, not uniformly distributed across the domain.

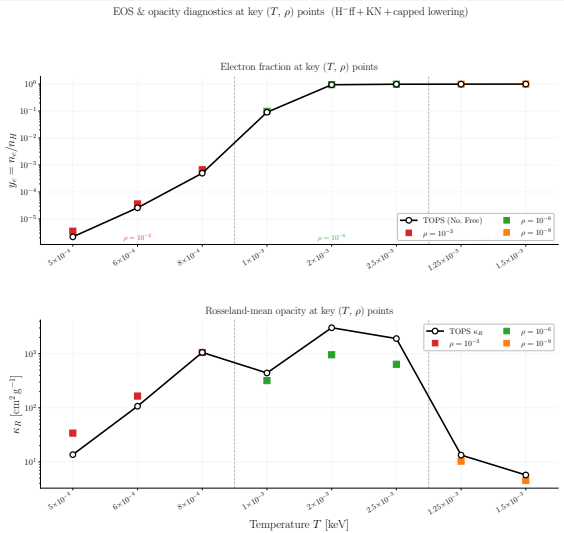
Cold Dense Corner: Why Full Lowering Is Too Crude



- **Full lowering** over-ionizes the cold dense corner by ∼16× in y_e, driving $\kappa_{\rm R}$ to ∼13× the TOPS value.
- **No lowering** avoids this pathology but under-corrects the partially neutral regime.
- **1 eV cap** gives the best benchmark compromise: y_e within 1.6× of TOPS at the coldest points.



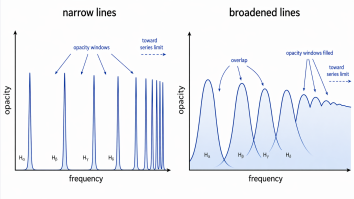
EOS Diagnostic: Is the Remaining Mismatch an Ionization Problem?

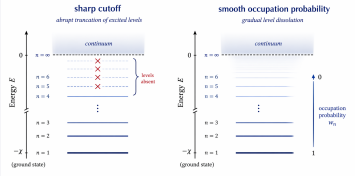


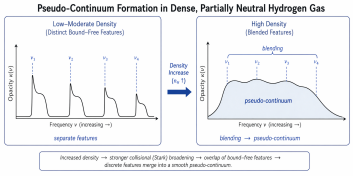
- At discrepant points, the electron fraction y_e is already **reasonably close** to the TOPS reference.
- Yet κ_R remains **significantly underestimated**.
- The residual discrepancy **cannot** be attributed to an EOS or ionization-balance failure.
- This points directly to **missing opacity physics**.

Reasonable y_e does not guarantee correct κ_R.

Physical Interpretation of the Remaining Discrepancies







Line broadening & overlap

Smooth occupation probability

Pseudo-continuum merging

Hot regime

- Bias removed by Klein–Nishina.
- <5% error above 10 mkeV.

Partially neutral ($\rho \sim 10^{-6}$ g cm^{-3})

- Persistent −30 to −50% bias.
- Missing Lyman/Balmer/Paschen lines, broadening, and blanketing.

Cold dense regime

- Missing bound-bound opacity.
- Smooth occupation probabilities & pseudo-continuum.

Conclusions

Benchmark Outcome

- **238 / 276** grid points within 10% of TOPS (86% overall).
- Near-perfect in the dilute regime and at $T \gtrsim 10$ mkeV.
- Dominant continuum physics already captured over most of the formal domain.

Key Physics Lessons

- H^- free-free is essential in the cold dense corner.
- Klein–Nishina correction is essential above ~3 keV.
- Full ionization-energy lowering is too crude; the 1 eV cap is a benchmark-calibrated phenomenological repair.

Remaining Open Issues

- **Dominant:** missing H bound-bound opacity (Lyman/Balmer lines, blanketing).
- **Dense regime:** smooth occupation probabilities and pseudo-continuum behavior not included.
- Full Hummer–Mihalas EOS is the natural next step.

A physically interpretable continuum-only model achieves 86% agreement with LANL/TOPS — the residual discrepancy is understood and physically attributed.

Full Production Physics Inventory

Included in production model

- ✓ Klein–Nishina electron scattering
- ✓ e–p free-free absorption (Gaunt factor corrected)
- ✓ Neutral-H bound-free photoionization (shells $n = 1 \ldots n_{\mathrm{cut}}$)
- ✓ H^- bound-free absorption (empirical fit)
- ✓ H^- free-free absorption (John 1988)
- ✓ Density-dependent level truncation
- ✓ Capped ionization-energy lowering (1 eV cap)

Not included / approximated

- × Bound-bound (line) opacity
- × Non-LTE effects
- × Full Hummer–Mihalas occupation probabilities
- × Full pressure-ionization / dense-plasma EOS
- × Quantum $g_{bf}(\nu)$ Gaunt factors (uses $g_{bf} = 1$)
- × Compton / Kompaneets energy redistribution

The production model is intentionally continuum-focused; omitted physics is used to interpret the residual discrepancies.

Klein–Nishina Scattering Implementation

$$\kappa_{\rm es}(\nu) = \frac{n_e\,\sigma_{\rm KN}(\nu)}{\rho}, \qquad \sigma_{\rm KN} \longrightarrow \sigma_T \quad (h\nu \ll m_e c^2)$$

$$\frac{\sigma_{\rm KN}}{\sigma_T} = 1 - 2x + \frac{26}{5}x^2 - \frac{133}{10}x^3 + \frac{1144}{35}x^4 + \cdots, \qquad x \equiv \frac{h\nu}{m_e c^2}$$

Numerical implementation

- Low-x Taylor branch used for $x < 0.05$ to maintain numerical stability near the Thomson limit.
- Full Klein–Nishina formula applied for $x \geq 0.05$; transition is continuous.
- Corrects the *total* scattering cross section only.

Scope and validation

- Does **not** include Compton energy exchange, Kompaneets evolution, or frequency redistribution.
- Regression tests verify: correct Taylor coefficients, branch continuity at $x = 0.05$, and monotone suppression relative to σ_T at high x.
- Reduces $\kappa_{\rm es}$ overestimate from ~15% to <4% at $T = 10$ keV.

Hydrogen Continuum Absorption

(1) e–p Free-Free Absorption

$$\alpha_{\rm ff} \propto n_e n_p T^{-1/2} \nu^{-3} \left(1 - e^{-h\nu/k_B T}\right) g_{\rm ff}(\nu, T)$$

- Kramers-like continuum absorption.
- Stimulated-emission corrected.
- Approximate quantum Gaunt factor $g_{\rm ff}(\nu, T)$ following Karzas&Latter1961, Hummer1988.
- Provides the baseline continuum opacity in the fully ionized regime.

(2) Neutral-H Bound-Free

$$\kappa_{\rm bf,H} = \frac{1}{\rho} \sum_n n_n \, \sigma_{n,\rm bf}(\nu) \left(1 - e^{-h\nu/k_B T}\right)$$

$$\chi_{n,\rm eff} = 13.6 \left[\frac{1}{n^2} - \frac{1}{\left(n_{\rm max}^{\rm phys}\right)^2}\right] \text{eV}$$

- Shell-by-shell hydrogenic treatment, summed over $n = 1 \ldots n_{\rm cut}$.
- Photoionization thresholds lowered using float $n_{\rm max}^{\rm phys}$ (not integer $n_{\rm cut}$).
- $g_{\rm bf} = 1$ adopted as a controlled approximation; quantum corrections neglected.
- Bound-bound line opacity is **not** included.

Negative-Hydrogen Continuum Opacity

(1) H^- Bound-Free

$$\sigma_{\mathrm{H}^-,\mathrm{bf}} = 1.53\times10^{-16}\,\lambda^3\left(\frac{1}{\lambda}-\frac{1}{\lambda_0}\right)^{3/2}\ \mathrm{cm}^2$$

$\lambda_0 = 1.64\ \mu$m; λ in μm.

- $h\nu \le 0.754$ eV: $\sigma = 0$ (below threshold)
- $0.754 < h\nu \le 10$ eV: empirical fit applied
- $h\nu > 10$ eV: $\sigma = 0$; no extrapolation
- Stimulated-emission corrected in opacity integral.

(2) H^- Free-Free (John 1988)

$$k_\lambda^{\mathrm{ff}} = 10^{-29}\sum_{j=1}^{6}\theta^{(j+1)/2}\left[A_j\lambda^2 + B_j + \frac{C_j}{\lambda} + \frac{D_j}{\lambda^2} + \frac{E_j}{\lambda^3} + \frac{F_j}{\lambda^4}\right]\ \mathrm{cm}^4\mathrm{dyne}^{-1}$$

$$\alpha_{\mathrm{ff},\mathrm{H}^-} = k_\lambda^{\mathrm{ff}}\,n_{H^0}\,P_e, \quad P_e = n_e k_B T$$

- $\theta = 5040/T$; λ in μm.
- Valid: $T \in [1400, 10080]$ K, $\lambda > 0.1823\ \mu$m.
- Returns **zero** outside validity domain.
- Stimulated-emission included in fit coefficients; no additional factor applied.

LTE EOS and Density-Dependent Level Truncation

Species Balance & Equilibrium

$$n_{H^0} + n_p + n_{H^-} = \frac{\rho}{m_H}, \quad n_p = n_e + n_{H^-}$$

$$\frac{n_e n_p}{n_{H^0}} = \left(\frac{2\pi m_e k_B T}{h^2}\right)^{3/2} \frac{2}{Z_H} e^{-\chi_{H,\mathrm{eff}}/k_B T}$$

$$\frac{n_{H^0}\, n_e}{n_{H^-}} = \frac{Z_H}{Z_{H^-}} \frac{g_e}{\lambda_{\mathrm{th,e}}^3} e^{-\chi_{H^-}/k_B T}$$

- EOS solved by a one-dimensional root-find in n_e at each (T, ρ).
- Four species: H^0, p, e^-, H^-; conservation enforced exactly.
- $\chi_{H,\mathrm{eff}}$ uses the capped lowering prescription (see next slide).

Density-Dependent Level Truncation

$$n_{\max}^{\mathrm{phys}} = 12 \left(\frac{n_H}{10^{15}\ \mathrm{cm}^{-3}}\right)^{-2/15}$$

$$n_{\mathrm{cut}} = \max\left[1,\ \min\left(n_{\max}^{\mathrm{user}},\ \lfloor n_{\max}^{\mathrm{phys}} \rfloor\right)\right]$$

- $n_{\max}^{\mathrm{phys}}$: **float** — used in all energy formulas and threshold lowering.
- n_{cut}: **integer** — used only in partition sums and level-population arrays.
- Separation avoids discrete jumps in energy quantities at density thresholds.
- Example: $n_{\mathrm{cut}} = 16$ at $\rho = 10^{-12}$; $n_{\mathrm{cut}} = 2$ at $10^{-3}\ \mathrm{g\,cm^{-3}}$.

Capped Ionization-Energy Lowering

$$\Delta\chi_{\text{raw}} = \frac{13.6}{\left(n_{\max}^{\text{phys}}\right)^2}\ \text{eV}$$

$$\Delta\chi_{\text{eff}} = \min(\Delta\chi_{\text{raw}},\ 1.0\ \text{eV})$$

$$\chi_{H,\text{eff}} = 13.6\ \text{eV} - \Delta\chi_{\text{eff}}$$

Prescription

- Float $n_{\max}^{\text{phys}}$ is used consistently in all energy-lowering formulas.
- The 1 eV cap prevents catastrophic over-ionization in the cold dense regime.
- Without the cap, full-strength lowering over-ionizes by $\sim 16\times$ relative to TOPS at the coldest benchmark points.

Interpretation

- The cap is a **phenomenological regularization** selected for stable benchmark behavior.
- It does not implement Hummer–Mihalas occupation probabilities or a self-consistent pressure-ionization model.
- Results in the cold dense corner should be interpreted accordingly.

Validation and Regression Tests

General Validation Checklist

- ✓ EOS positivity and species conservation
- ✓ Opacity component non-negativity (all 5 mechanisms)
- ✓ Threshold and domain enforcement for H bf and H^- fits
- ✓ Rosseland mean positivity and finiteness
- ✓ x-grid convergence (base vs. 4× refined; <1% mid-domain)
- ✓ No spectral oscillations or numerical spikes
- ✓ Regression consistency at selected (T, ρ) benchmark points

Klein–Nishina Regression Tests

- ✓ Low-x Taylor coefficients verified to all four terms
- ✓ Branch continuity at $x = 0.05$
- ✓ Monotone suppression relative to σ_T at high x
- ✓ Thomson limit recovered as $x \to 0$

Total: 161 / 161 tests passing.

All validation checks pass on the final production build.

238 / 276 within 10% (final production model, validation-verified)

Scientific Scope and Non-Claims

This model claims

- ✓ Controlled LTE continuum-opacity calculation over the formal benchmark domain.
- ✓ Self-consistent species balance for H^0, p, e^-, H^-.
- ✓ Good agreement over most of the LANL/TOPS benchmark domain (86%).
- ✓ Physically localized and regime-specific residual discrepancies.
- ✓ Clear identification of the likely missing opacity channels.

This model does not claim

- × Complete reproduction of TOPS.
- × Inclusion of bound-bound line opacity.
- × Non-LTE validity.
- × First-principles non-ideal EOS.
- × Full Hummer–Mihalas occupation probabilities.
- × Compton / Kompaneets energy redistribution.
- × Full dense-plasma pressure-ionization physics.

The value of the model is that its approximations are explicit and its residuals are physically diagnosable.